Extended reading — Week 13: Multi-Agent RL and AlphaZero

May 21, 2026

1. Yu, Velu, Vinitsky, Wang, Bayen, Wu (2022) — "The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games"

NeurIPS 2022. arXiv:2103.01955.

Summary. MAPPO — PPO with a centralised critic and parameter-shared actor — matches or beats QMIX, MADDPG, and COMA across the StarCraft Multi-Agent Challenge, MPE, and the Hanabi benchmark. The paper carefully ablates five design choices (value normalisation, observation normalisation, gradient clipping, PPO clip range, advantage normalisation) and ships a recipe that just works.

Why it matters this week. Until 2021 the cooperative-MARL community had effectively abandoned vanilla policy gradients — value-decomposition methods (QMIX, VDN) dominated. MAPPO showed the abandonment was premature; the right hyperparameters and a global critic recover everything. This is now the default cooperative-MARL algorithm.

Key passage.

> "We find that, with simple modifications, PPO achieves strong results in [...] cooperative multi-agent games. *In contrast to the common belief that on-policy methods are sample-inefficient*, our results show that MAPPO is competitive with or superior to off-policy methods, while requiring fewer hyperparameter tweaks."

Commentary. The italics are the heart of the contribution. The conventional wisdom about on-policy methods being sample-inefficient came from single-agent benchmarks; in MARL the off-policy methods' inability to handle non-stationarity costs them more than PPO's on-policy data hunger.

2. Silver, Hubert, Schrittwieser et al. (2018) — "A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play"

Science 362, 1140–1144. (AlphaZero paper.)

Summary. A single algorithm — self-play + MCTS + a two-headed network + the policy-iteration view — trained from scratch on chess, shogi, and Go, beat the strongest hand-engineered programs in each. No human game knowledge except the rules.

Why it matters this week. AlphaZero is the cleanest demonstration of *learning* an evaluation function from *search*. The MCTS visit distribution is a policy-improvement operator; training the network to match it converts that improvement into amortised inference, which then powers the next search. This is policy iteration with two function approximators (the prior π_θ and the value v_θ).

Key passage.

> "Each MCTS uses the neural network f_θ to guide its simulations. [...] The output of the search are the visit-count probabilities π over the root state. The neural network is then trained by self-play reinforcement learning: it plays games against itself with parameters θ, and is updated [...] to make p more closely match π and the predicted value v more closely match the game outcome z."

Commentary. "Match π" is the policy-improvement step — the visit-count distribution is strictly better at the root than π_θ because of MCTS lookahead. "Match z" is the policy-evaluation step. Iterate, and you have generalised policy iteration with both pieces amortised into a network.

3. Lowe, Wu, Tamar, Harb, Abbeel, Mordatch (2017) — "Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments"

NeurIPS 2017. arXiv:1706.02275. (MADDPG.)

Summary. The original CTDE actor-critic. Each agent has its own actor (local obs only) and its own critic that takes the global state *and all agents' actions* as input. The critic is centralised; the actor is decentralised. Solves cooperative, competitive, and mixed environments where independent DDPG fails.

Why it matters this week. MADDPG defined the CTDE recipe MAPPO would later refine. Reading it makes clear what MAPPO inherited (centralised critic, joint training) and what it changed (PPO surrogate replaces DDPG's deterministic gradient; parameter sharing replaces per-agent critics).

Key passage.

> "We propose a centralised critic that has access to the observations and actions of all agents [...] this avoids the non-stationarity problem [...] from the perspective of any individual agent, the environment appears non-stationary as the other agents' policies change."

Commentary. This single sentence diagnoses the core MARL problem (non-stationarity) and prescribes the cure (centralised information at training time). Everything in cooperative deep MARL since 2017 builds on it.

4. Wang, Gleave, Tu, Mei et al. (2023) — "Adversarial Policies Beat Superhuman Go AIs"

ICML 2023. arXiv:2211.00241.

Summary. KataGo — the strongest open-source Go bot, comfortably super-human in standard play — can be defeated by an adversarial policy that exploits a narrow tactical pattern (a "cyclic group" attack). Crucially, the adversarial policy is itself weak in normal play; the exploit lives entirely in a region of state-space that self-play never visited.

Why it matters this week. This is the single most important paper for *any* safety-conscious deployment of self-play. The lesson is that self-play certifies in-distribution skill, not robustness. The same blind spot is what makes RLHF reward models vulnerable to jailbreaks: optimisation pressure against the training distribution doesn't guarantee performance off-distribution.

Key passage.

> "Despite KataGo's strength in standard play, our adversaries achieve >97% win-rate against KataGo at superhuman strength settings. The adversaries do not exhibit strong play in general — they are not capable of beating a beginner-level human player at standard Go — they simply exploit a specific weakness in KataGo's policy."

Commentary. "They do not exhibit strong play in general" is the part to internalise. The next time someone says "but our self-play agent is super-human", ask: *at what?* If you only measure in-distribution win-rate you have not measured robustness.

5. Lanctot, Zambaldi, Gruslys, Lazaridou, Tuyls, Pérolat, Silver, Graepel (2017) — "A Unified Game-Theoretic Approach to Multi-Agent Reinforcement Learning"

NeurIPS 2017. arXiv:1711.00832. (PSRO.)

Summary. PSRO frames MARL as iterative best-response: maintain a population of policies, at each iteration solve a meta-game (uniform or Nash) over the current population, train a best-response against the meta-strategy, add it to the population. Generalises double-oracle and fictitious play.

Why it matters this week. For mixed-motive or large-action games where single-pair self-play diverges, PSRO and its descendants (α-*PSRO*, "Joint Policy Space Response Oracles", AlphaStar's league) are the practical answer. The PSRO loop is also the cleanest framework for thinking about "train a robust agent" — robustness is what falls out of best-response against a diverse population.

Key passage.

> "The Policy-Space Response Oracle (PSRO) framework generalises classic game-theoretic solution concepts to large or continuous strategy spaces. [...] PSRO can recover fictitious play, independent learning, and the double-oracle algorithm as special cases."

Commentary. The unification is the point. Once you see fictitious play, independent learning, and double-oracle as the three faces of one operator, you can reason about which population-update rule matters for your problem (uniform for robustness, Nash for exploitability minimisation, α-rank for evolutionary stability).

Reading exercises

1. Read Yu 2022 §4 and identify which of the five MAPPO design choices most affects performance on the SMAC benchmark.
2. Sketch the policy-improvement / policy-evaluation analogy between AlphaZero and the policy-iteration algorithm of Week 2. Where does MCTS sit in the diagram?
3. Re-read the Wang 2023 abstract and design a stress test you would add to AlphaZero training on Tic-Tac-Toe to surface the same pathology. (Hint: train a "parasite" policy in parallel.)
4. PSRO: write down the meta-game payoff matrix for a 2-pure-policy population in zero-sum, and pick a meta-solver (uniform vs Nash) for each of: robustness, exploitability minimisation, exploration.

Forward connections

Multi-agent thinking returns repeatedly in the rest of the curriculum.

- Week 17 (exploration + meta-RL) revisits PSRO in the context of population-based exploration.
- Week 19 (DPO zoo) is implicitly multi-agent: the language model is one player, the human evaluator a second, the reward model an imperfect referee.
- Week 24 (capstone) returns to the two-sided matching market with formal CMDP constraints.

Sourced from `research/deeprl_core_sota.md` (§Multi-agent) and `research/safety_rl_sota.md` (§Self-play overfit + adversarial policies).